Logos Experiment 16: Large Scaling + Extended Training

Experiment 16 extends the large 124M-parameter Logos model from 5000 to 7500 training iterations. The purpose is to test whether the best large-scale recipe continues improving without changing architecture, parameter count, checkpoint size, or context length.

1. Experiment Objective

The objective was to evaluate whether the 124M-parameter Weight Tying + SwiGLU decoder-only model was still undertrained after Experiment 15. Since Experiment 15 achieved its best validation loss at the final iteration, Experiment 16 increased the training budget while keeping the model and data setup unchanged.

2. Configuration

Component	Setting
Run name	logos_exp16_large_scaling_weight_tying_swiglu_extended_train
Model type	Decoder-only GPT-style causal language model
Dataset	OpenWebText subset, 100,000 samples
Tokenizer	GPT-2 tokenizer
Layers	12
Embedding dimension	768
Attention heads	12
Head dimension	64
Context length	512 tokens
Batch size	32
Tokens per step	16,384
FFN type	SwiGLU
Weight tying	Enabled
Training iterations	7500
Optimizer	AdamW
Learning rate	3e-4 with warmup and cosine decay

3. Final Results

Metric	Value
Parameters	124,031,232
Best iteration	7500
Best validation loss	4.2301
Final validation loss	4.2254
Final validation perplexity	68.40
Checkpoint size	473.20 MB
Peak VRAM	17.73 GB
Training time	1736.33 sec = 28.94 min
Generation speed	83.90 tokens/sec
Generation latency	11.92 ms/token
Average GPU power	292.93 W
Estimated GPU energy	141.11 Wh

4. Comparison with Experiment 15

Metric	Exp 15: 5000 iters	Exp 16: 7500 iters	Change
Parameters	124.03M	124.03M	No change
Dataset	100k	100k	No change
Context	512	512	No change
Best val loss	4.4690	4.2301	Improved
Final val loss	4.4663	4.2254	Improved
Perplexity	87.03	68.40	Improved

Metric	Exp 15: 5000 iters	Exp 16: 7500 iters	Change
Checkpoint	473.20 MB	473.20 MB	No change
Peak VRAM	17.73 GB	17.73 GB	No change
Training time	19.80 min	28.94 min	Higher
Energy	96.03 Wh	141.11 Wh	Higher

5. Scientific Interpretation

Experiment 16 is a major improvement over Experiment 15. Extending training from 5000 to 7500 iterations reduced validation loss from 4.4663 to 4.2254 and perplexity from 87.03 to 68.40. This improvement occurred without increasing model size, checkpoint size, or VRAM usage.

The tradeoff is higher training time and energy consumption, which is expected because the model was trained for a longer duration. The result indicates that the 124M Logos model was still learning after 5000 iterations and that longer training remains beneficial under the current setup.

6. Conclusion

Experiment 16 is currently the best Logos model overall. It provides the lowest validation loss and perplexity achieved so far while keeping the same architecture as Experiment 15. The best current model is therefore a 124.03M-parameter decoder-only Transformer using Weight Tying and SwiGLU, trained for 7500 iterations on 100k OpenWebText samples.